Predicting the Zero-Shot Transfer of a Promptable Video Tracker

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Abstract

Conservation teams increasingly monitor wildlife with camera traps, which record far more footage than anyone can watch. Locating each animal in a video and following it from frame to frame is the bottleneck, and a new generation of promptable trackers promises to remove it: name an animal in ordinary words and the model finds and follows every one that appears, without ever having been trained on that species. SAM 3 is the current example. The difficulty is that this ability is uneven in ways nobody can anticipate — strong on one animal or location, poor on the next — so a team pointing such a model at a new species has no way of judging beforehand whether to trust its output or fall back on manual annotation.

This dissertation asks whether that judgement can be made in advance. It tests a natural intuition: that a model should struggle most on the animals and places least like those it has seen before. Were that true, one could measure how unfamiliar a species or a setting is — cheaply, without labelling anything and without running the model at all — and use the measurement to forecast reliability. Four such measures of unfamiliarity are tested, and the question is put separately to two distinct abilities: finding the animal at all, and keeping the identities straight once it has been found.

The answer is that it does not work. On a large camera-trap benchmark of wild animals, no measure of unfamiliarity predicts performance on species or places the model has not seen. The only property that predicts anything is how large the animal appears in frame — a nuisance rather than a discovery — and a second attempt based on how difficult the footage looks, such as darkness and visual clutter, collapses into that same size effect. The finding is not an accident of one dataset or one model: it survives on a second, independent dataset and on two further trackers of quite different design. Nor is it a failure of method, since the same test reliably detects the size effect whenever it is present.

The contribution is therefore a carefully bounded negative result, the first for this class of model, and with it a practical caution: resemblance to familiar training data is no guarantee that such a tracker will work, and reliability has to be checked rather than assumed.

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List of Abbreviations

`pHOTA`	Promptable Higher Order Tracking Accuracy — the benchmark's tracking score
`pDetA`	Its *detection* half: were the right animals found? (our primary target)
`pAssA`	Its *association* half: was each identity kept over time?
HOTA	Higher Order Tracking Accuracy, the metric `pHOTA` derives from
IoU	Intersection over Union, the mask/box overlap measure
GLM	Generalised Linear Model (here a support-weighted logit-link regression)
CI	Confidence Interval (throughout, a 95% group-cluster-bootstrap interval)
MAE	Mean Absolute Error
ΔMAE	Out-of-sample MAE gain over a mean predictor; positive = better than guessing the mean
LSO / LLO	Leave-Species-Out / Leave-Location-Out cross-validation
VIF	Variance Inflation Factor, a collinearity diagnostic
LMG	Lindeman–Merenda–Gold, the dominance-analysis variance split
MMD	Maximum Mean Discrepancy, a distributional distance
OOD	Out-Of-Distribution
ATC	Average Thresholded Confidence, an after-running accuracy estimator
GT	Ground Truth
IR	Infrared (night-time camera-trap imagery)
RLE	Run-Length Encoding, the COCO mask format
LVIS	Large Vocabulary Instance Segmentation, the class vocabulary BURST uses

Chapter 1

Introduction

1.1 Motivation

Finding an animal in a video and following it over time is a need shared by several fields. Ecologists use it to survey wild populations from camera-trap footage. Neuroscientists and behavioural biologists use it to quantify what an animal does frame by frame in the laboratory, where the tracked trajectory is often the measurement the science rests on, and where a whole tool-chain now exists to recover pose and turn it into behavioural labels [17, 20, 22]. The bottleneck is the same in both cases: video is cheap to record and expensive to annotate.

This dissertation works on the wild side of that divide, where the problem is hardest. Camera traps have become a standard tool in wildlife monitoring, and a single survey leaves a team with far more footage than anyone can watch. Somebody still has to work out which animal is in each clip, how many there are, and what they do. Done by hand this is slow and expensive, and it is the step that limits how much ecology gets done at all.

Automating it means solving two problems, not one. The first is *finding* the animal in a frame. The second is *following* it from frame to frame, so the same individual keeps the same identity for as long as it is visible. Only the second yields the counts and behavioural sequences both communities want: in a study of social behaviour, losing an identity means losing the very animal whose behaviour was being measured. It is also the harder of the two.

Neither is easy on real footage. Animals are camouflaged against the background they live in, appear at any distance and viewpoint, and are cut in half by vegetation. Much of the footage is recorded at night in infrared. And because a camera trap is fixed, the same scene repeats in every clip from that site, which gives a model an easy shortcut: learn the background rather than the animal [5, 32]. Section 2.2 sets these difficulties out in full.

For years the standard answer was to train a detector on a fixed list of species and deploy it. It works where it was trained. It degrades sharply at camera sites and regions it has not seen [3, 15]. And adding a species that was not on the list means collecting and labelling more data — precisely the cost a team opening a new site is trying to avoid.

Promptable models change that arrangement, because the prompt can now be a phrase. Give SAM 3 [7] `"impala"` and it finds, segments and follows every impala in the clip, having never been trained on that species. A team can point a tracker at an animal it has never seen and get an answer straight away.

That is the appeal. The liability is that the answer is not equally good everywhere. Zero-shot performance varies sharply from one animal to the next, and from one place to the next. No rule tells a practitioner, before running the model, whether its output can be trusted for a given animal in a given place. The choice left is an unattractive one: trust the output blindly, or fall back on the manual annotation the model was meant to replace.

This dissertation studies that gap. It asks whether such a rule exists — first for SAM 3, and then whether whatever we find is specific to it.

1.2 Problem statement and research question

The appeal of a zero-shot model is that it skips the labelling step. That is also what makes it hard to trust. The normal way to check whether a model works on your data is to label some of it and measure. But a team that could afford to do that would have had less need for a zero-shot model in the first place. So these tools get deployed precisely where nobody has checked that they work.

The problem is a general one. Any foundation model applied to a case nobody validated sits in the same position. The capability is broad, the guarantees are absent, and the user has no principled basis for deciding how far to trust the output. What would settle it is a rule that can be checked *in advance*: some property of the target, cheap to measure, that says whether the model will cope.

There is an obvious candidate, and practitioners already reason with it informally — distance from the familiar. A model ought to do worse on things unlike what it has seen. If that intuition held quantitatively, it would be genuinely useful. You could measure how far a new species or place sits from the model's reference. That number would then act as a deployment gate, applied before spending compute and before annotating anything.

This dissertation asks whether that rule exists. Precisely: can a promptable tracker's zero-shot accuracy on an unseen (species, place) be predicted in advance, from properties measurable without running the model or scoring its output? And which factors, if any, govern that transfer?

We ask it separately for the two halves of the task, because tracking is built on detection. The model must first find the animal, then hold onto its identity from frame to frame. The two can fail for different reasons. Finding is about whether the animal is recognisable at all; following is about whether the scene lets an identity survive. A single accuracy number would average them together and hide the answer, so we keep them apart throughout.

Asking the question at all requires a demanding dataset. It needs species labels structured enough to say how far apart two species are, plus location metadata, timestamps, and dense per-frame tracking annotation — all in one corpus. SA-FARI [29] carries all four, which is why we use it; Section 2.7 sets out its contents.

We study SAM 3 on SA-FARI as the primary case. We then ask whether the answer is specific to it. The analysis is repeated on a second, independent dataset, and two other promptable trackers are swapped in on the same data (Section 4.11).

1.3 Hypotheses

The research question asks two things at once: *whether* transfer can be predicted, and *what governs* it. The hypotheses below turn that into something a dataset can answer. Each names a factor, commits to a direction, and can be rejected by the evidence.

Two terms are needed first, because the rest of the dissertation rests on them.

The tracker is scored with a standard tracking metric that splits cleanly into two halves (Section 2.3). **`pDetA`**, detection accuracy, scores the finding half: were the right animals located? **`pAssA`**, association accuracy, scores the following half: did each animal keep one consistent identity for as long as it was visible? Both run from 0 to 1, and we model them separately for the reason given in Section 1.2.

A predictor is **label-free** if computing it needs no annotation of the target footage and no measurement of how the tracker performed on it. Only such a predictor can be evaluated *before* the model is run, which is exactly what a deployment gate would require. The four tested here are all *distances*: how far a target species or place sits from a fixed reference set, measured by taxonomy, by appearance, by the surrounding scene, and by time.

H0 (null)

The label-free distances carry no predictive information about `pDetA` or `pAssA`. Transfer is not distance-driven.

H1 (detection)
`pDetA` falls as the species becomes more novel — further from the reference in taxonomy and in appearance.

H2 (association)
`pAssA` falls as the scene becomes harder — an unfamiliar background, night or infrared capture, visual clutter.

H1 and H2 divide the research question between the two halves of the task, and each attaches the half to the factor most likely to govern it. H1 is the one that matters most in deployment. If it held, a team could look up an unfamiliar species and know in advance whether detection is likely to hold up. H2 rests on a different intuition. Holding an identity across frames should depend on the scene rather than on which animal it is. If so, the difficulties catalogued in Section 2.2 — darkness, clutter, an unfamiliar background — ought to bite here rather than on detection.

H0 is what remains if both fail. It is not an absence of an answer but a substantive one. It would mean the distance intuition, however reasonable it sounds, does not survive contact with the data — and that a deployment gate cannot be built this way.

1.4 Contributions

To our knowledge this is the first study to ask whether a promptable video tracker's zero-shot transfer can be predicted before the model is run, from properties that require no labelling of the target. It is also the first to put that question to an out-of-sample test. Section 2.8 places that claim against the existing literature. What the work contributes is the following.

(i) **A testable formulation of the question, and a protocol for answering it.** Turning "will this model work on my data?" into something that can be falsified takes a specific design. The frozen tracker is scored with the official metric. Every predictor is measured against a reference fixed in advance, so that no information about the target can leak in. And validation holds out whole species and whole locations rather than individual clips. That protocol is not tied to the predictors tested here. Any future before-running estimator can be held to the same bar, which is what makes the negative answer useful rather than merely discouraging.

(ii) **The detection–association split brought to label-free evaluation.** Work of this kind has so far predicted a single accuracy number. Asking separately whether *finding* and *following* an animal transfer for different reasons is new, and it is what allows an answer to be attributed to one half of the task instead of averaged across both.

(iii) **The answer, established at scope.** Label-free distance does not predict transfer. The finding is reported as a bounded negative result. It is also shown not to be an artefact of one dataset or one model, surviving an independent second dataset and two architecturally different trackers (Chapter 4).

(iv) **An account of why the obvious candidate fails, and the caution that follows.** The one apparent effect reduces to how large the animal appears in the frame. That is a warning for future work: any study measuring visual novelty across species has to control for apparent size, or it will rediscover the same artefact. It is also a caution for practice. Proximity to familiar training data is not a safety guarantee, and reliability still has to be checked rather than assumed.

1.5 Dissertation outline

Chapter 2 reviews the related work and states precisely the gap this dissertation fills. Chapter 3 describes the data and the two splits, how transfer is measured, the four label-free distances, and the statistical procedure used to test them. Chapter 4 reports the results, including the out-of-sample validation, the robustness checks, and the replications on a further dataset and two further trackers. Chapter 5 interprets those results, reconciles them with the prior literature, and sets out the limitations and threats to validity together with the ethical considerations. Chapter 6 concludes and identifies what is left open for future work.

Chapter 2

Background and Related Work

This chapter reviews the work this dissertation builds on. It covers the class of model under study, the difficulties that make animal detection and tracking hard in the first place, how tracking is scored, the literature on estimating a model's performance without labels, what is known about distribution shift in camera-trap vision, the representations that make a distance between species meaningful, and the datasets involved. It closes by stating the gap that remains.

2.1 Promptable and open-vocabulary segmentation and tracking

For most of the last decade, detection and segmentation models were trained against a fixed list of categories. Such a model cannot be asked about anything outside that list. Extending it to a new category means collecting labelled examples and training again. In camera-trap work the standard tool of this kind is MegaDetector [4], a detector trained on millions of camera-trap images to localise animals, people and vehicles, and widely deployed as a first-pass filter over survey footage.

The *Segment Anything* line broke the fixed-list assumption. It replaced the label set with a prompt: a user indicates what should be segmented, and the model returns a mask for it. SAM 2 [24] carried the formulation from images to video, propagating a mask across frames from a point or box placed in one of them.

SAM 3 [7] changes what the prompt may be. Instead of a point or a box it accepts a short noun phrase, and returns every instance matching that phrase, segmented and tracked across the clip. The category need never have been seen in training. This is a zero-shot, open-vocabulary detect–segment–track capability, and SAM 3 is the current state of the art for it.

Other models reach open-vocabulary video understanding by different routes. GLEE [30] trains a single unified model to detect, segment and track objects at scale, driven by a text query. Florence-2 [31] takes a caption-style approach, using one sequence-to-sequence model across a range of vision tasks and emitting regions in response to a task prompt, though without a confidence score attached to them. These are the main alternative designs in the same space.

2.2 What makes animals hard to detect and track

Wild footage is an unusually hostile input, and the difficulties are worth setting out because they are what any tracker deployed on it must survive.

The first is that animals are often hard to see at all. Many are camouflaged against the habitat they live in, and the resulting low contrast between object and background defeats methods that assume a visually salient foreground. This is a recognised problem in its own right: camouflaged object detection is studied as a separate task with dedicated benchmarks, precisely because standard detectors underperform on it [11].

The second is variability of appearance. A camera trap is triggered by whatever passes, so an animal may fill the frame in one clip and occupy a few dozen pixels in the next. It may face the lens or present only a flank, and vegetation frequently occludes part of it. A bounding box is a poor description of an animal in such conditions, which is one reason segmentation — labelling the pixels belonging to the animal — is the more informative output.

The third is image quality. Much wildlife footage is captured at night under infrared illumination, where colour information is absent and contrast is poor. Infrared and thermal imaging are established wildlife monitoring modalities with their own detection characteristics, and their properties differ enough from daylight imagery that observation strategies are designed around them [6]. Motion blur is common, since animals are typically moving when they trigger the camera.

The fourth is context. A camera trap is fixed, so the same scene recurs in every clip from that site, and a model can come to rely on the scene rather than on the animal. This failure mode has a literature of its own, reviewed in Section 2.5.

The fifth applies to tracking rather than detection. Keeping an identity attached to the right animal is a distinct problem from finding it. Animals leave the frame and re-enter it, pass behind vegetation, and in groups may be near-indistinguishable from one another. Identity errors are treated as a first-class failure mode in multi-object tracking benchmarks, which annotate visibility explicitly and report identity switches separately from misses [9]. Recovering the identity of a specific individual across sightings is hard enough in wildlife imagery to constitute its own research area [26].

2.3 Evaluating multi-object tracking

Multi-object tracking is scored by comparing predicted tracks against annotated ones. The difficulty is that two very different failures — missing the object, and swapping its identity — both reduce a single accuracy figure, and earlier metrics combined them in ways that were hard to interpret.

Higher Order Tracking Accuracy (HOTA) [16] was introduced to separate them. At each localisation threshold α it forms a detection term (`DetA`: were the right objects found?) and an association term (`AssA`: were identities kept consistent over time?), then combines them as a geometric mean:

$$\mathrm{HOTA}_\alpha = \sqrt{\mathrm{DetA}_\alpha \cdot \mathrm{AssA}_\alpha}, \qquad \mathrm{pHOTA} = \frac{1}{|\mathcal{A}|} \sum_{\alpha \in \mathcal{A}} \mathrm{HOTA}_\alpha, \tag{2.1}$$

averaged over the threshold set $\mathcal{A} = \{0.05, 0.10, \ldots, 0.95\}$. Because the two terms are formed separately before being combined, a HOTA score can be read either as a single number or as two.

SA-FARI scores promptable tracking with `pHOTA`, a promptable variant that decomposes identically into `pDetA` and `pAssA`.

2.4 Predicting model performance without labels

A now-mature literature shows that a model's performance under distribution shift can be anticipated without labelling the target data.

Miller et al. [19] observe that out-of-distribution (OOD) accuracy is strongly linearly correlated with in-distribution accuracy, across many models and many shifts. Baek et al. [2] show that the agreement between a pair of models tracks OOD accuracy closely enough to estimate it from unlabelled data alone. A parallel strand reads the model's own statistics on the unlabelled target. Average thresholded confidence [12] calibrates a confidence threshold on source data and reads off target accuracy as the mass above it, while Deng and Zheng [10] regress classifier accuracy directly on a distribution-shift feature — the Fréchet distance between train and test feature sets.

The idea has only recently reached structured outputs. Yoo et al. [33] estimate object-detection performance without ground truth, from the spatial consistency and confidence reliability of pre-NMS boxes. Reverse classification accuracy [28] predicts per-image segmentation quality by training a reverse model against a labelled reference.

Two properties characterise essentially all of this work. First, the target is an image classifier scored by a single scalar, or at most a detector or segmenter operating on a static image. None of it concerns identity maintained over time, and none decomposes the estimate into separate components. Second, every one of these methods runs the model on the target and inspects what comes out — its predictions, its confidences, or its agreement with another model. Each estimates the score of a model that has already been executed.

2.5 Distribution shift in camera-trap and wildlife vision

That camera-trap models degrade at new locations is well established. *Terra Incognita* [3] showed that recognisers excellent at trained camera sites fail sharply at unseen ones. WILDS [15] consolidated this, via iWildCam, as a canonical real-world shift. PanAf-FGBG [5] demonstrated that behaviour-correlated backgrounds causally drive the drop, and proposed a latent-space mitigation.

The claim that scene context, rather than the object, drives such failures has a longer pedigree in general vision. Xiao et al. [32] isolate foreground from background on ImageNet, in the ImageNet-9 "Background Challenge", and show that a classifier scores well above chance from the background *alone*. An adversarially chosen background flips a correctly classified foreground up to 87.5% of the time. Background is genuine signal, not noise. Rosenfeld et al. [25] make the parallel point for detectors: transplanting an object into an unfamiliar scene suppresses its detection, flips its predicted identity as it moves, and disturbs other detections in the image, exposing a reliance on context and co-occurrence.

Xiao et al. [32] also report that more accurate and more robust models rely on background *less*, which suggests the effect is not fixed but a property of how strong the model is.

What this literature shares is its direction of travel. It measures degradation after the fact, or intervenes on the input to mitigate it. It does not forecast per-site or per-species reliability in advance. It also studies classifiers and behaviour recognisers rather than promptable trackers, scored by a single accuracy with no detection/association split.

2.6 Representations of species and scenes

Measuring how far one species sits from another, or one scene from another, requires a representation in which such a distance means something.

Vision foundation models turn out to carry that structure. BioCLIP [27] shows that a model trained on a large biological image corpus encodes taxonomic structure across the tree of life, its feature space reflecting the hierarchy that runs from kingdom down to species. Appearance and phylogeny are correlated, so a representation that separates species visually will tend also to order them taxonomically.

General-purpose encoders offer the same service without biological supervision. DINOv2 [21] learns visual features self-supervised, and they transfer across domains without fine-tuning. CLIP [23] aligns images with natural-language descriptions, producing a space in which images and text share coordinates. Both are widely used as off-the-shelf feature extractors.

2.7 Datasets

Two datasets are central to this work.

SA-FARI. SA-FARI [29] is a large open multi-animal camera-trap tracking dataset, released alongside SAM 3. It contains 11,609 densely annotated videos spanning 99 species, recorded at 741 locations across four continents. Every species carries a seven-level biological taxonomy running from kingdom to species, and every video carries a creation timestamp. Annotations are per-frame masks with persistent identities, and the benchmark ships an official evaluator, `VEval`.

Two features set it apart. The taxonomy makes the relationship between two species measurable rather than a matter of judgement, and the location and timestamp metadata do the same for place and era. The benchmark also includes abundant *hard negatives*: prompts naming a species that is absent from the clip, which a correct run should answer with nothing. These make it possible to study false alarms alongside misses.

SA-FARI ships a train/test partition drawn along locations, so the test camera sites are unseen while most species occur on both sides of it.

BURST. BURST [1] is a video benchmark for object recognition, segmentation and tracking, built on the TAO [8] video corpus. It annotates 482 categories from the LVIS [13] vocabulary with per-frame masks and persistent track identities, and a portion of that vocabulary is animal categories. Its footage comes from internet, driving and film sources rather than camera traps, and it carries neither location nor timestamp metadata. It is mask-native and considerably more varied in subject matter than SA-FARI, which makes it a useful independent point of comparison.

2.8 Summary and gap

Predictable OOD generalisation is established for classifiers scored by a scalar [2, 10, 12, 19] and, very recently, for static detectors and segmenters [28, 33]. Camera-trap vision has thoroughly documented, but not forecast, location-driven degradation [3, 5, 15]. Most tellingly, a recent unified study of camera-trap recognition over time [14] explicitly names "predicting when zero-shot models will succeed at a new site" as an open question, and stops at naming it.

No prior work predicts, *before running the model* and from label-free distances, the zero-shot transfer of a promptable video tracker. None separates that prediction into detection and association. That is the gap this dissertation addresses.

Chapter 3

Data and Methodology

3.1 Research design and evaluation criteria

This work is a predictive-modelling study: we ask whether label-free distances measurable in advance (X) predict a frozen tracker's zero-shot performance ($Y = \texttt{pDetA}/\texttt{pAssA}$), and which factors govern that transfer for finding versus following an animal. Because the target is a bounded score and the observations are grouped by species and location, the analysis is a weighted regression with group-aware, out-of-sample validation rather than a tuned tracker.

Definition of success and the decision rule. The intended deliverable is a validated, out-of-sample predictor of `pDetA/pAssA` from label-free distances, with confidence intervals that respect the group structure. A distance is taken to predict transfer only if it satisfies both criteria:

(i) its coefficient's group-cluster-bootstrap 95% confidence interval (CI) excludes zero; and

(ii) the grouped cross-validated model beats a mean-predictor baseline by a margin that is significant under a paired group-bootstrap.

H1 (detection ← species novelty) and **H2** (association ← environment difficulty) are supported only when their respective distances meet this bar on the split that isolates them; if neither does under these deliberately powerful tests, we conclude **H0** — distance explains little variance — which is itself a valid result and motivates the representational-probing direction of Chapter 5.

3.2 Dataset, splits, and the analysis unit

We use SA-FARI [29] throughout. In its annotations, each test is a probe — a (video, text-prompt) pair; the prompt resolves the animal's species and its seven-level taxonomy, while each video's location and timestamp give place and time. We aggregate scores to a cell — a (species, location, time) group, keyed by a stable numeric species identifier — and attach support counts (n_{frames}, n_{masklets}, n_{videos}) used for weighting and as covariates. Hard negatives (prompts for absent species) are kept, as they drive the false-positive side of detection; the one exception is the stratified per-species cap used to make the large pooled pass feasible, under which hard negatives are set aside from inference and analysed instead by the separate false-positive pass (Section 3.7).

Two complementary splits. Inspection of the annotations shows the shipped split is disjoint by location but shares species across train and test (Section 2.7); on that split species-distance is ≈ 0 for every cell, so it cannot test species novelty. Because SAM 3 is frozen and zero-shot on all of SA-FARI, the split only defines our reference distribution, so we are free to repartition the same videos. We therefore use two near-orthogonal experiments and run inference only once (scoring is per-video and split-agnostic):

Split A — species hold-out (primary).
Pooled species are partitioned by a leave-species-out view, so a held-out species' distance is to the nearest other species; species-distance varies while the environment stays largely familiar, isolating H1. Validated by leave-species-out cross-validation.

Split B — location hold-out (secondary).
The official train/test split, whose unseen locations isolate H2 and whose benchmark comparability carries the initial score-sanity check.

Split A is a constructed hold-out — legitimate because the model is frozen, and reported as such. At the scale we work with, the location split (Split B) yields 1,447 scored cells over 113 taxonomy-bearing category identities (99 species) and 92 unseen test locations: 346 are positive cells — the queried species is present, so `pDetA`/`pAssA` are defined — and 1,101 are hard-negative cells, drawn from 1,486 hard-negative probes. The species hold-out (Split A) pools the present species of both official splits into 1,025 scored cells across the 99 species.

3.3 Measuring transfer

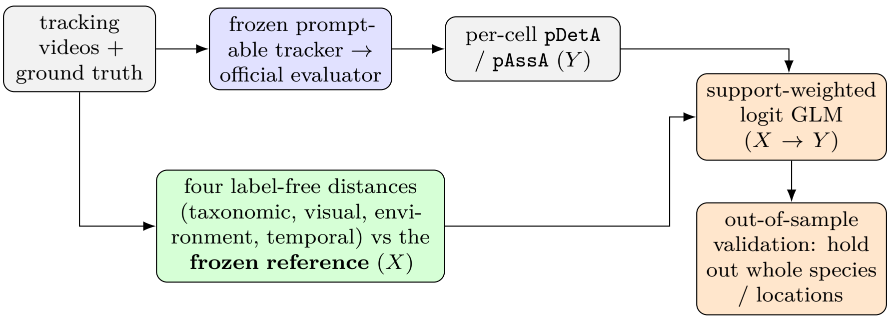

Figure 3.1: The measurement-and-modelling pipeline, identical for every dataset and tracker. *Top:* the frozen tracker is scored by the official evaluator into per-cell detection/association scores (the target Y). *Bottom:* four label-free distances are computed against a frozen reference (the predictors X). A support-weighted logit GLM relates X to Y, validated out of sample by holding out whole species and locations. Only the GLM is fitted; the tracker is never trained.

We now instantiate this pipeline on the primary case — SAM 3 on SA-FARI — whose concrete choices, numbers, and benchmark status the rest of this section reports; the external replications and the model swap (Sections 3.8–3.8.1) then re-run the *same* pipeline on other datasets and trackers. SAM 3 [7] is run frozen over the probe videos — species-specific prompts as the primary condition, a generic "animal" prompt for robustness — and scored with the official `VEval` evaluator, which we never re-implement. This yields a per-cell `pDetA`/`pAssA`/`pHOTA` [16] with its support. A single inference pass over the union of the Split A and Split B probes serves both experiments. Predictions are stored per (video, species) so that the several species-prompts issued on one video do not collide; the harness is resumable at per-video granularity and never updates a weight.

We aggregate the frame-level scores two ways and keep the two distinct. The micro score is the dataset-level `VEval` figure — the published SA-FARI operating point, ≈0.65 mask-`pHOTA` — which we report only to confirm we reproduce the benchmark. The macro score is the per-cell, support-weighted mean over positive cells, and it is our modelling target: the model is fit cell-by-cell, so the cell is the unit that must carry the score. Because the macro mean is a positives-only per-cell average it sits lower (≈0.53); every measurement number we quote is the macro value, labelled as such.

No retraining; validity of the frozen protocol. One point is worth stating plainly, because it is easily misread: **SAM 3 is never retrained or fine-tuned in this work.** The only object we fit is the small support-weighted logistic regression that maps the label-free distances X onto the observed scores Y; the tracker is the public checkpoint, run once, its weights untouched. Reusing that one inference across both splits is therefore sound rather than a leak — because the model is frozen, a cell's score cannot depend on which reference partition we later overlay to compute its distances.

This also settles the benchmark's status. SA-FARI is a held-out, zero-shot benchmark for the released SAM 3: the SAM 3 report [7] does not list SA-FARI among its training sources, and the SA-FARI paper [29] reports the released checkpoint zero-shot (the $\approx$0.65 mask-`pHOTA` ballpark we reproduce in measurement) alongside separately fine-tuned rows whose gain we deliberately forgo. We can therefore verify non-membership at the dataset level; we cannot verify it frame-by-frame, since SAM 3's full pretraining corpus is undisclosed. We do not claim otherwise: every distance is measured against the SA-FARI train reference, not against SAM 3's true (unknown) pretraining distribution, and the familiarity proxy (Section 3.5.6) is the hedge for exactly this gap.

3.4 The frozen reference (leakage firewall)

For each split the reference ("seen") species, locations, and taxonomy are frozen once; every distance is then computed against that reference only, so no probe information can enter a feature — verified by a unit test that swaps the reference and checks the distances move as expected. We assert the axis that is genuinely disjoint (location for Split B; the held-out species for Split A) and report any residual species overlap rather than asserting it away.

3.5 Label-free distance features

All distances are computed against the frozen reference and require no measurement of the tracker's performance on the target — no `pDetA`, no `pAssA`, no annotation of how well SAM 3 did. That is the sense in which they are *label-free*, and it is the sense the research question needs: they are available *before* the model is run. One qualification belongs here rather than in a footnote. As implemented on an annotated benchmark, the visual, environment and size features read the dataset's ground-truth masks to separate animal from scene. That is a convenience of working on SA-FARI, not a requirement of the method — in deployment the same species prototype would be built from the handful of exemplar images a practitioner already has for a named species — but it does bound the claim, and we return to it in Section 5.5.

3.5.1 Taxonomic distance

Lowest-common-ancestor tree steps from the probe's species to the nearest reference species (shared genus = 1, family = 2, order = 3, …); the leave-species-out reference keeps this non-trivial for shared species.

3.5.2 Visual distance

Each species is fingerprinted by a prototype — the re-normalised mean DINOv2 [21] embedding of its ground-truth-mask-cropped animals — and the visual distance is the cosine gap from a probe species' prototype to that of the nearest other reference species (leave-species-out on Split A), with a CLIP [23] encoder as a robustness check. Using the dataset's ground-truth masks means this feature needs no SAM 3 (Figure 3.2b).

3.5.3 Environment distance

Embedding distance between the animal-masked-out background scene and the nearest reference location, plus interpretable scene covariates from the same frames: a day/night–infrared (IR) flag and a low-light fraction (both from per-channel colour statistics of the frame), and a clutter score given by the variance of a Laplacian over the masked background — a cheap measure of local edge/texture density, not a colour statistic (the animal-erased scene is Figure 3.2c).

(a) frame + ground-truth mask (b) visual: mask-cropped animal (c) environment: animal erased

Figure 3.2: The two image-based distances on a camera-trap frame (a collared peccary). **(a)** ground-truth mask (orange); **(b)** the mask-cropped animal the visual distance fingerprints; **(c)** the animal mean-inpainted, which the environment distance fingerprints. None of these features runs SAM 3.

3.5.4 Temporal gap

The minimum absolute year gap between a cell's capture year (from the per-video creation-datetime metadata) and the nearest reference year. A usable timestamp is present for only 32 of 1447 cells (2.2%), and every one of those 32 has a gap of exactly 0 — zero variance — so the temporal axis is reported as untested rather than as a tested null.

3.5.5 Animal size (confound control)

Mean log ground-truth mask area per species. This is not a hypothesis variable but a nuisance covariate, included to separate genuine visual novelty from the fact that visually-distinctive species tend to be large and thus easier to segment; its role in exposing the one apparent effect as an artefact is Section 3.6.

3.5.6 SAM 3 familiarity proxy

Because SAM 3's true pretraining corpus is undisclosed, every distance above is measured against the SA-FARI reference, not against what SAM 3 actually saw (Section 3.9 caveat). The familiarity proxy hedges this by reading familiarity from the model itself: how separable each species is in SAM 3's own vision-encoder feature space. The same mask-cropped animals used for the visual distance are pushed through SAM 3's image encoder, mean-pooling the patch-token features to a 1024-d vector, and each species is scored three ways — silhouette (own-cluster compactness vs the nearest other species), nearest-prototype (the visual distance recomputed with SAM 3's encoder rather than DINOv2), and Mahalanobis typicality against the reference feature Gaussian. Like the other distances it is before-running (an image property, not a function of SAM 3's tracking output, so it is non-circular) and label-free. Each metric is validated at the exact leave-species-out bar (Bonferroni over the three), size-controlled, and put in a head-to-head against the DINOv2 visual distance: the decisive question is whether it adds any out-of-sample gain over the visual distance + size (Section 4.8).

3.6 Statistical modelling and hypothesis testing

3.6.1 Per-target regression

Each bounded target (`pDetA`, `pAssA`) is modelled separately with a support-weighted logit-link (fractional-logit) regression over the standardised distances plus the scene covariates and a $\log(n_{\text{frames}})$ covariate, so that "rare" is never mistaken for "far":

$$\operatorname{logit} \mathbb{E}[y] = x^\top \beta, \qquad \text{fit with variance weights } w = n_{\text{frames}}, \tag{3.1}$$

with $y \in [0, 1]$ the cell's score and x the standardised design. Detection and association are analysed apart because their decomposition is the intellectual core of the project. In practice association is a near-degenerate target on this data — most cells contain a single object, so `pAssA` tracks `pDetA` almost exactly — and it is therefore reported as such rather than as an independently powered test (Chapter 5).

3.6.2 Honest, group-aware inference

Naive model confidence intervals are invalid here for two reasons: support-weighting by frame count inflates the effective sample size, and the distances are constant within a species or a location, so the hundreds of cells are really only tens of independent groups (pseudo-replication). We therefore report group-cluster-bootstrap intervals — resampling whole species and whole locations, refitting, and taking percentile intervals — as the honest confidence interval for every coefficient.

3.6.3 Out-of-sample validation and the significance test

The fitted models are validated out of sample by holding out whole groups (leave-species-out on Split A, leave-location-out on Split B) and predicting the held-out group. The out-of-sample gain over a mean-predictor baseline is tested with a paired group-bootstrap that resamples whole held-out groups, yielding a confidence interval and a one-sided p-value on the gain — the criterion (ii) of Section 3.1.

3.6.4 Variance attribution and confound control

Variance is attributed to each factor by dominance/commonality analysis with a VIF collinearity check — not by four univariate fits — and raw correlations are reported as a model-free cross-check. Where a distance reaches significance, we refit with the animal-size covariate to test whether the effect is a size confound rather than novelty.

3.6.5 The intrinsic-difficulty pivot and the nested decomposition

Because the sole apparent effect proved to be a size confound, we ask a complementary question: is transfer governed by the intrinsic difficulty of the target imagery rather than its novelty? We promote the label-free scene properties already measured for the environment axis — the low-light fraction, the clutter score, and the night/IR flag — from nuisance covariates to predictors of interest, and test them against the same out-of-sample bar. To separate a genuine difficulty effect from the size confound, we fit a nested sequence of models — support only; size only; difficulty only (no size); difficulty plus size — and compare their out-of-sample gains, so that any apparent difficulty effect that is really size is exposed as such (Chapter 4, Table 4.4). We also report each scene property's cell-level correlation alongside its location-level value, because aggregation inflates correlations (the ecological-correlation fallacy).

Detecting a signal, not asserting its absence. These tests are demonstrably able to detect a real effect: the animal-size covariate not only has an interval excluding zero but validates out of sample at the same whole-group hold-out bar the distances fail (Chapter 4), and the estimation and cross-validation machinery is unit-tested to recover planted effects on synthetic data. A null under a test that fires for size is therefore evidence of *absence of signal*, not of *absence of power*.

3.6.6 Design-robustness checks

To show the null is not an artefact of a single design choice, we re-run the entire hold-out procedure with one component of the pipeline swapped at a time: the visual encoder (DINOv2 → CLIP), the crop (mask-cropped animal → a whole-frame bounding box that keeps the background), the prompt (species-specific → a generic "animal"), and the visual-distance functional itself (nearest-prototype cosine → a distributional Fréchet/MMD distance between the target and reference embedding sets, after AutoEval [10]). Each variant is fitted and validated at the identical whole-group hold-out bar of Section 3.1, with significance judged after a Bonferroni correction over the variant family; the outcomes are reported in Chapter 4 (Table 4.11).

3.7 Detection precision: false positives on hard negatives

The `pDetA` model conditions on the animal being present, so it measures recall-given-present, not hallucination. On hard negatives (the queried species is absent) a correct run returns no masklet; we read the prediction directly, flag any returned masklet above a score threshold as a false positive, attach each species' taxonomic and visual distance, and test — at the species level, to avoid pseudo-replication — whether the hallucination rate rises with novelty.

3.8 Independent replication on an external dataset

Every SA-FARI result reuses one frozen inference pass on one dataset, so the two splits are near-orthogonal but not statistically independent. To test whether the before-running null is a property of the transfer problem rather than of SA-FARI, we re-run the entire pipeline — unchanged frozen SAM 3, the same official scorer, the same distances, and the same leave-species-out cross-validation and paired group bootstrap — on an independent tracking dataset. The abstraction boundary that makes this cheap is the annotation schema, not the loader: an offline adapter converts the dataset into the SA-Co annotation JSON (the SA-FARI/VEval schema) the loader and evaluator already read, so nothing downstream changes. Only the small logit GLM is ever fitted; SAM 3 is frozen throughout.

BURST (mask-native, many-species). The animal subset of the BURST validation split [1] (built on TAO [8]): 41 animal species across 132 video × class cells, with native COCO-RLE masks scored by mask-HOTA. The animal categories are selected from BURST's 482 LVIS classes [13] by a WordNet filter [18] (hyponyms of animal.n.01) and mapped to a seven-level biological taxonomy where resolvable (25/41 species full, the rest left undefined). BURST carries no site or timestamp metadata, so the cell key is the per-video sequence and leave-species-out is the only valid grouping; the association target is genuinely non-degenerate here, so both components are tested. Masks stream from the RWTH server and TAO frames from the Hugging Face mirror by HTTP range, so only the annotated frames of the capped animal videos are fetched.

Interpreting the replication. BURST is held to the exact same out-of-sample bar as SA-FARI, with the animal-size confound as the positive control — it validates out-of-sample on SA-FARI, so a test that stays flat for the distances there is genuinely finding no signal; where it too fails to fire (as on BURST) the design is simply too small to certify power, marking the result as consistency rather than confirmation. We report convergence, not independent proof: BURST is better powered by cell count than the SA-FARI location split but still cannot exclude a small before-running distance effect (Section 4.9).

3.8.1 The controlled model swap

The dataset-level analysis varies the data while holding the tracker fixed; to test whether the null is a property of SAM 3 specifically we also vary the tracker. A controlled model swap holds everything else constant — the same cells, the same ground truth, the same four label-free distances, and the same support-weighted GLM under leave-species-out cross-validation — and substitutes SAM 3 with a second open-vocabulary, text-promptable tracker; only the target `pDetA` changes, since the distances are functions of the data and ground truth, not of the tracker. This is a controlled swap, not an independent replication: it reuses the same cells and ground truth, so it isolates the tracker as the sole varying factor. We swap in two challengers of deliberately different architecture — GLEE [30], a single unified detect–segment–track model, and Florence-2 [31] + SAM 2 [24], a caption-style box generator with no confidence score followed by a mask-propagating tracker.

Two safeguards keep the swap fair. First, a measurement gate: before fitting any regression on a challenger's scores we require its per-cell `pDetA` to be non-degenerate (real spread across cells, not floored at zero); a regression on degenerate scores has no variance to explain and would report a spurious "null," so a challenger that scores ≈ 0 on a dataset is dropped from the swap on that dataset rather than modelled. Second, scoring fairness: because open-vocabulary detectors' confidence scores are calibrated to their own training domain, we verify that any near-zero challenger score is a genuine detection failure and not an artefact of the evaluator's fixed confidence gate — by re-scoring under a box-based metric and under a threshold-free average-precision metric with no gate at all. This gate is why the swap runs on BURST, not SA-FARI: SA-FARI is SAM 3's own co-released benchmark and its camera-trap domain collapses the challengers' calibration even where they localise well, so it cannot yield a fair, non-degenerate challenger score, whereas SAM 3 scores no lower on the independent BURST than on SA-FARI. Every challenger configuration is fixed in advance from the model's own defaults, run once, and reported as-is (Section 4.11).

3.9 Reproducibility, implementation, and constraints

The model is frozen and scored by the official evaluator; the reference (seen) set is frozen before any distance is computed; hard negatives are retained; and all inference is single-GPU and inference-only. The pipeline is config-driven (every constraint is a toggle, so an ablation flips config, not code) and covered by hermetic tests; every reported number is regenerated from the repository, and split definitions, seeds, and reference sets are persisted so both experiments are reproducible end to end. The engineering safeguards that keep the measurement sound — memory-lean feature assembly, per-(video, species) prediction keying, trimming the ground truth to the scored videos before evaluation, and the size-confound control — are implemented in the pipeline and covered by the test suite, each framed as a validity safeguard.

Chapter 4

Results and Evaluation

4.1 Measurement

As an initial score-sanity check, zero-shot SAM 3 on the SA-FARI test split reproduces the published SAM 3 ballpark: the dataset-level (micro) `VEval` score is ≈ 0.65 mask-`pHOTA`, matching the operating point reported with the benchmark. Table 4.1 reports a different and lower quantity — the per-cell, support-weighted (macro) mean over positive cells, which is the target we model. It sits at ≈ 0.53 because it averages positives-only within each cell rather than pooling over the whole dataset (Section 3.3); the two are not in conflict.

Table 4.1: Zero-shot SAM 3 on the SA-FARI test split — support-weighted mean over 346 scored cells (of 1447).

Metric (mask)	Value
pDetA	0.527
pAssA	0.546
pHOTA	0.533

4.2 Fitted coefficients

This section reports the first of the two criteria a distance must clear to count as predictive (Section 3.1): does its coefficient's confidence interval exclude zero? Table 4.2 gives the standardised logit-GLM coefficients on the location split (Split B — the same 346 positive cells as Table 4.1); the species hold-out is reported separately in Section 4.5.

Two conventions make the table readable. The coefficients are *standardised*: each is the change in the logit of the score for a one-standard-deviation increase in that factor, so magnitudes are comparable down the column even though the raw factors are in different units — tree steps, cosine gaps, a binary flag. A *negative* coefficient means the score falls as the factor rises, which is the direction H1 and H2 predict. The intervals are not the model's own: because a distance is constant within a species or a location, the 346 cells behave like only a few tens of independent groups, and an interval computed as though every cell were independent would be far too narrow. These are group-cluster-bootstrap intervals that resample whole species and whole locations (Section 3.6.2), so they are wide deliberately — that width is the real uncertainty, not a defect of the data.

Read that way the table says one thing: no factor clears the bar. Every interval spans zero, for detection and association alike, so criterion (i) already fails — before the out-of-sample test of Section 4.4 is even reached. Two details deserve reading rather than skipping. First, the point estimates for taxonomic and visual distance are negative here, nominally the direction H1 predicts; but the intervals are far too wide for the sign to carry information, and on the species hold-out — where species novelty actually varies — the same two coefficients come back *positive* (Section 4.5), the opposite of what H1 says. Second, taxonomic distance carries by far the widest interval ($[-3.87, +0.09]$ for detection), and that is a property of this split rather than a finding about taxonomy: the official split shares species across train and test, so taxonomic distance is ≈ 0 for almost every cell here and the coefficient is estimated from almost no variation. It is measured imprecisely, not measured and refuted, and we count it as neither. Testing species novelty properly is precisely what the constructed species hold-out exists for.

Table 4.2: Standardised logit-GLM coefficients (group-cluster-bootstrap 95% CIs) for detection and association. Every distance's CI spans zero.

Factor	pDetA coef. [CI]	pAssA coef. [CI]
Taxonomic distance	-0.362 [-3.87, 0.09]	-0.368 [-3.40, 0.08]
Visual distance	-0.144 [-0.50, 0.14]	-0.122 [-0.49, 0.18]
Environment distance	0.042 [-0.19, 0.24]	0.013 [-0.23, 0.21]
Clutter	-0.066 [-0.38, 0.29]	-0.087 [-0.41, 0.28]
Night / IR	-0.405 [-0.94, 0.11]	-0.462 [-1.02, 0.11]
log(support)	-0.216 [-0.51, 0.17]	-0.224 [-0.53, 0.18]

4.3 How much the distances explain, and the difficulty pivot

We first ask how much of the variation in detection the four distances account for. A dominance analysis (the LMG/Shapley method, which fairly splits shared explanatory power between correlated predictors rather than double-counting it) shows they account for almost nothing: together the four explain only $\approx 3\%$ of the variation in detection, and no single distance stands out (Table 4.3). A natural worry is that the distances might be so correlated with one another that a real effect is hidden by them cancelling out; the VIF column rules this out (VIF, the variance inflation factor, measures how entangled each predictor is with the others — a value near 1 means "essentially independent"). Every VIF is ≈ 1, so the 3% is genuinely all there is, not a masked signal. The one distance that looked predictive (visual distance) turned out to be

Table 4.3: Variance partition for detection: each distance's unique share of R^2 (LMG/Shapley) and its VIF. The four distances explain only $\approx 3\%$ of the variance and every VIF ≈ 1 — the null is not masked by collinearity.

Factor	unique R^2	share	VIF
Taxonomic	0.014	48%	1.06
Visual	0.012	39%	1.06
Environment	0.004	13%	1.00
Temporal	0.000	0%	–

explained by animal size — visually distinctive species tend to be larger, and larger animals are easier to detect, so "novelty" was really "size" in disguise (a confound). This prompts a different question: rather than how novel the animal is, does how hard the scene is predict detection? We therefore test three label-free scene properties — low light, visual clutter, and a night/infrared flag, each measured directly from the frames with no SAM 3 run and no target label — against the same bar (Table 4.4). Bundled together with animal size, these difficulty signals appear to work (they beat the baseline by $\approx +0.017$ on both hold-outs). But when we separate the contributions, the entire gain comes from size: size *on its own* carries it, while low light *on* *its own* does not beat the baseline and adds almost nothing once size is accounted for. The apparent night/IR effect is also inflated by looking at it the wrong way: measured per location the correlation with detection looks moderate ($r = -0.377$), but that is an averaging artefact — measured at the level of the individual cell, where the model actually operates, it shrinks to -0.13, and clutter's is 0.00. So the difficulty idea is **also null**: the only label-free property that predicts detection is animal size.

Table 4.4: Nested size decomposition of the intrinsic-difficulty test (detection, 346 cells). Entries are the out-of-sample MAE gain over a mean baseline, Δ [95% CI] (p), under leave-species-out and leave-location-out. The entire gain sits in the size covariate; low-light *alone* does not validate — the difficulty signals pass only by riding on size.

Model	Leave-species-out Δ [CI] (p)	Leave-location-out Δ [CI] (p)
log(support) only	+0.001 [-0.007, +0.008] (0.432)	+0.002 [-0.005, +0.008] (0.291)
log(area) only (size)	+0.015 [+0.002, +0.028] (0.014)	+0.017 [+0.009, +0.025] (0.001)
Low-light only (no size)	+0.003 [-0.006, +0.011] (0.233)	+0.004 [-0.003, +0.011] (0.123)
Low-light + size	+0.017 [+0.004, +0.032] (0.011)	+0.020 [+0.011, +0.030] (0.000)

4.4 The headline test: predicting unseen species and locations

This is the decisive test. We train the model on some species and locations, then ask it to predict detection for held-out ones it never saw, and compare its error to the simplest possible guess — always predicting the average score (Table 4.5). The result is a **null**: on both hold-out schemes (hold out whole species; hold out whole locations) and for both targets, the distance model does no better than that trivial average. The improvement in error over the baseline (ΔMAE, the drop in mean absolute error) is within noise of zero in every case, and none is statistically significant: the predictions collapse onto the mean rather than tracking the truth (Figure 4.1).

The null holds up because the test demonstrably works. Run unchanged on the same cases, it *does* recover a real effect: animal size beats the baseline out of sample at the identical bar, significant on both schemes (ΔMAE = +0.015 leave-species-out, +0.017 leave-location-out; Table 4.4). A test that fires for size while staying flat for the distances tells us the distances genuinely carry *no* predictive signal — the flat result is an absence of signal, and the size result is what confirms it.

Table 4.5: Out-of-sample error vs a mean predictor on the **location split** (Split B; the 346 positive cells). Each row is one cross-validation grouping scheme — leave-species-out or leave-location-out — applied within these same cells; Δ is the MAE gain over baseline, with a paired-bootstrap 95% CI and one-sided p. No scheme's gain is significant.

CV scheme	Target	n	MAE	Baseline	Δ [95% CI]	p
location	pAssA	346	0.289	0.295	+0.007 [-0.005, +0.018]	0.13
location	pDetA	346	0.293	0.298	+0.004 [-0.007, +0.015]	0.23
species	pAssA	346	0.289	0.295	+0.006 [-0.006, +0.017]	0.16
species	pDetA	346	0.293	0.298	+0.004 [-0.006, +0.015]	0.24

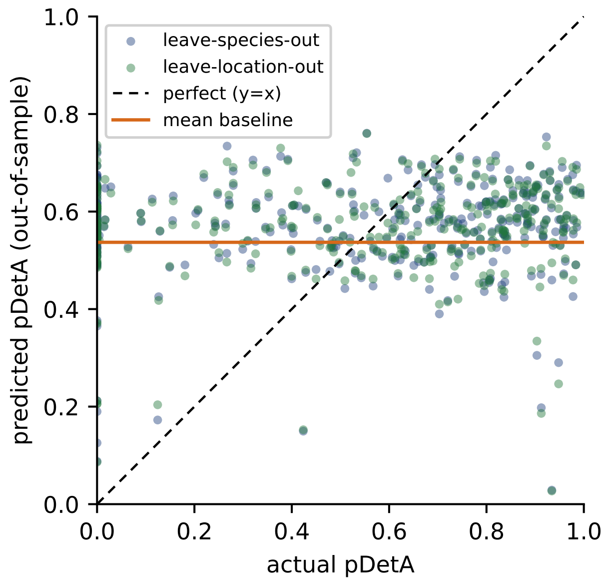

Figure 4.1: Out-of-sample detection predictions vs the truth, held-out species and locations. They collapse onto the mean baseline (orange) instead of the diagonal (dashed) — the distances carry no predictive power.

4.5 Species hold-out (H1): novelty and the size confound

The primary novelty test — leave-one-species-out over the pooled train+test species (99 species, 1025 scored cells) — is in Tables 4.6–4.8. Out-of-sample prediction is not significant on any hold-out (Table 4.8). The only coefficient whose CI excludes zero, visual distance (Table 4.7), is positive — meaning more-novel species are detected *better*, the reverse of what H1 predicts — and is a **confound**: visually-distinctive species are larger (visual $\leftrightarrow$ log-area $r = 0.22$) and larger animals score higher (log-area $\leftrightarrow$ pDetA $r = 0.30$, the strongest raw correlate of the score). Adding an animal-size covariate shrinks the visual coefficient from +0.37 to +0.22 (CI now spanning zero) while size itself is the significant driver, so "visual novelty helps" was really "bigger animals are easier". Taxonomic novelty is flat ($r \approx 0.00$). Novelty does not predict transfer.

Table 4.6: Zero-shot SAM 3 on the pooled train+test set — support-weighted mean over 1025 scored cells (of 2126).

Metric (mask)	Value
pDetA	0.536
pAssA	0.597
pHOTA	0.559

Table 4.7: Standardised logit-GLM coefficients (group-cluster-bootstrap 95% CIs) for detection and association, species hold-out.

Factor	pDetA coef. [CI]	pAssA coef. [CI]
Taxonomic distance	0.202 [-0.09, 0.43]	0.209 [-0.06, 0.42]
Visual distance	0.369 [0.11, 0.67]	0.271 [0.04, 0.54]
Environment distance	0.043 [-0.15, 0.17]	0.023 [-0.15, 0.13]
Clutter	-0.284 [-0.52, 0.02]	-0.230 [-0.46, 0.06]
Night / IR	-0.286 [-0.77, 0.22]	-0.439 [-0.92, 0.06]
log(support)	-0.020 [-0.14, 0.16]	0.051 [-0.06, 0.23]

Table 4.8: Out-of-sample error vs a mean predictor on the **species hold-out** (Split A; the 1025 pooled train+test cells over 99 species). Each row is one cross-validation grouping scheme — leave-species-out or leave-location-out — applied within these same cells; Δ is the MAE gain over baseline, with a paired-bootstrap 95% CI and one-sided p. No scheme's gain is significant. (Distinct experiment from Table 4.5, hence the different n and values.)

CV scheme	Target	n	MAE	Baseline	Δ [95% CI]	p
location	pAssA	1025	0.270	0.270	-0.000 [-0.006, +0.005]	0.52
location	pDetA	1025	0.274	0.274	-0.001 [-0.007, +0.006]	0.58
species	pAssA	1025	0.273	0.270	-0.003 [-0.014, +0.008]	0.74
species	pDetA	1025	0.277	0.274	-0.004 [-0.015, +0.009]	0.74

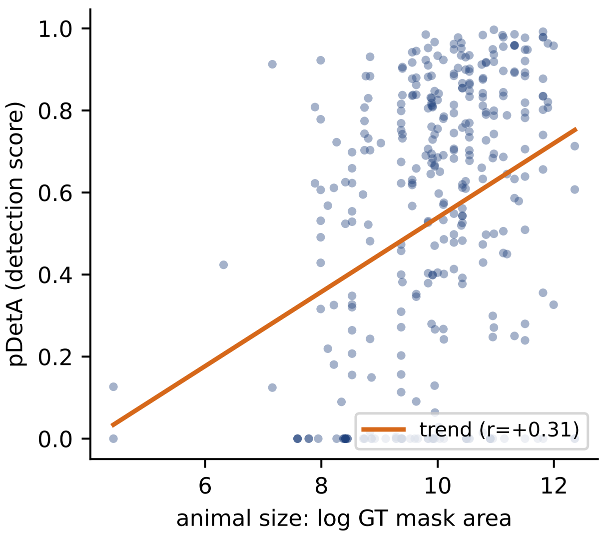

Figure 4.2: The one label-free property that predicts detection: animal size. Per-cell `pDetA` rises with log mask area ($r = +0.31$) — the confound behind the apparent visual-distance effect.

4.6 False positives / hallucination

The detection score conditions on the animal being present, so it measures recall, not hallucination. The 1486 hard negatives (queries for absent species) ask the complementary question, and are the closest the study comes to a validated before-running signal. SAM 3 returns a masklet anyway about **9.8**% of the time. This hallucination rate does *not* rise with taxonomic novelty ($r \approx +0.06$, n.s.); it falls with visual distinctiveness ($r \approx -0.33$), and this survives an animal-size control (partial -0.37) — a genuine, size-independent, label-free correlate, though its direction ("distinctive = safer") is the *opposite* of "novelty hurts". Crucially it is a **correlate, not a predictor**: predicting a held-out species' hallucination rate from visual distance beats a mean baseline by only +0.008 MAE and *just misses* out-of-sample significance ($p = 0.053$; Table 4.9). Even the study's best signal does not clear the bar.

Table 4.9: False positives on the 1486 hard negatives (113 species): overall hallucination rate 9.8%. More visually distinctive species hallucinate *less* — a real, size-independent correlate — but it does not validate out of sample (leave-species-out $p = 0.053$): a correlate, not a predictor.

Test (species-level)	statistic	95% CI	verdict
FP rate vs taxonomic distance	+0.06	[−0.17, +0.23]	n.s.
FP rate vs **visual distance**	−0.33	[−0.43, −0.23]	significant
visual, size-controlled (partial)	−0.37	[−0.47, −0.28]	significant
FP rate vs animal size	+0.13	[−0.06, +0.31]	n.s.
leave-species-out OOS gain	+0.008	[−0.002, +0.017]	$p = 0.053$

4.7 Ablations and robustness

Factor ablation. Refitting detection with each distance dropped in turn barely moves the fit or the (null) out-of-sample gain (Table 4.10): no single distance carries signal. The only design that changes anything is adding the animal-size covariate, which alone lifts the out-of-sample gain to significance (ΔMAE $\approx +0.02$, $p \leq 0.02$) — again consistent with size being the sole real correlate. Removing the support covariate does not rescue a distance effect either. The variance partition (Table 4.3) tells the same story from the other direction.

Table 4.10: Factor ablation for detection: McFadden pseudo-R^2 and the out-of-sample MAE gain over a mean baseline under leave-species-out (LSO) and leave-location-out (LLO). Dropping any distance barely changes the fit; only the animal-size covariate moves the score.

Design	pseudo-R^2	LSO Δ (p)	LLO Δ (p)
full (4 distances)	0.055	+0.004 (p=0.24)	+0.004 (p=0.23)
drop taxonomic	0.039	+0.000 (p=0.47)	+0.000 (p=0.49)
drop temporal	0.055	+0.004 (p=0.24)	+0.004 (p=0.23)
drop visual	0.048	+0.004 (p=0.22)	+0.004 (p=0.28)
drop environment	0.055	+0.006 (p=0.15)	+0.005 (p=0.17)
+ animal size	0.119	+0.018 (p=0.02)	+0.020 (p=0.00)
− support covariate	0.035	−0.002 (p=0.72)	−0.003 (p=0.73)

Design robustness. The null is not an artefact of the encoder, the crop, the prompt, or the distance functional (Table 4.11). Re-running the full leave-species-out and leave-location-out procedure with a CLIP rather than DINOv2 encoder, a whole-frame rather than mask-cropped embedding, and a distributional Fréchet or MMD rather than nearest-prototype distance each leaves the out-of-sample gain indistinguishable from the baseline (all ΔMAE ≤ 0.007, $p \geq 0.10$, every 95% CI spanning zero). A generic "animal" prompt is included for completeness: prompting non-specifically collapses per-species detection to a near-constant, mostly-zero target (mean `pDetA`≈ 0.14, 73% of cells exactly zero), so the distance model cannot beat — and, fitting a degenerate target, underperforms — a mean predictor. Across every variant the label-free distances fail identically, confirming the null is a property of the transfer problem, not of one modelling choice.

Table 4.11: Design-robustness sweep (detection). Each row re-runs the primary model with one design choice swapped; entries are the out-of-sample MAE gain, Δ [95% CI] (p), under leave-species-out (LSO) and leave-location-out (LLO) (Bonferroni $m = 5$). The null holds throughout — no encoder, crop, prompt, or distributional distance beats the mean.

Variant	LSO Δ [CI] (p)	LLO Δ [CI] (p)	Beats mean?
baseline (DINOv2, mask-crop, species prompt)	+0.004 [-0.006, +0.015] (0.237)	+0.004 [-0.007, +0.015] (0.235)	n.s.
CLIP encoder	+0.004 [-0.006, +0.014] (0.247)	+0.004 [-0.007, +0.014] (0.270)	n.s.
whole-frame (bbox + background)	+0.002 [-0.008, +0.013] (0.349)	+0.002 [-0.008, +0.013] (0.357)	n.s.
generic "animal" prompt	-0.346 [-0.381, -0.309] (1.000)	-0.343 [-0.363, -0.326] (1.000)	n.s.
Frechet distributional distance	+0.007 [-0.005, +0.019] (0.120)	+0.007 [-0.004, +0.019] (0.102)	n.s.
MMD distributional distance	+0.002 [-0.008, +0.012] (0.374)	+0.002 [-0.009, +0.012] (0.349)	n.s.

4.8 SAM 3 familiarity proxy

The distances measure novelty against the SA-FARI reference, not against SAM 3's undisclosed pretraining set. The familiarity proxy (§3.5.6) closes that gap by reading familiarity from SAM 3's own vision-encoder feature space — three separability metrics on the 346 positive cells (Table 4.12). The result is another null: on the leave-species-out novelty test none of the three clears the Bonferroni threshold ($\alpha = 0.017$) — the strongest, silhouette, reaches only $p = 0.034$ (ΔMAE $= +0.012$), and nearest-prototype and Mahalanobis do not come close ($p = 0.068$, 0.085). All three are dominated by the DINOv2 visual distance they re-encode ($r = 0.63$–0.85), and in the head-to-head none adds out-of-sample gain over the visual distance + size. The one curiosity is that silhouette, unlike the visual distance, is *not* a size artefact ($r_{\text{size}} = -0.16$) — it is the size-independent half of visual distinctness — yet it still fails to predict transfer.

The metrics do reach significance under leave-location-out ($p \leq 0.01$), but this is not a novelty test: the familiarity value is species-constant (within-species variance = 0), and on the location split the held-out locations' species are shared with the training locations, so a species-constant feature is never tested on an unseen species. Crucially, on these same 346 cells the before-running animal-size control does validate out-of-sample, so the leave-species-out familiarity null reflects a genuine absence of signal, not a lack of power. The null replicates on BURST (all three metrics $p > 0.13$ except a matching silhouette near-miss at $p = 0.020$; none adds over visual + size). Reading familiarity directly from SAM 3's representations therefore does *not* rescue before-running prediction — it recovers the same size-entangled visual-distinctness signal that was already H0, and closes the "but you do not know SAM 3's true training set" loophole in the null.

Table 4.12: SAM 3 familiarity proxy (detection): does a species' separability in SAM 3's own feature space predict `pDetA`? Each metric is size-controlled; ΔMAE (p) is the out-of-sample gain over a mean predictor. None clears Bonferroni ($\alpha = 0.017$) on either dataset, and all three re-encode the visual distance ($r_{\text{vis}} = 0.63$–0.85). As the size control validates on the same cells, this is a genuine null, not an underpowered one.

Metric	SA-FARI ΔMAE (p)	BURST ΔMAE (p)	r_{vis}	r_{size}	clears bar?
silhouette	+0.012 (0.034)	+0.012 (0.020)	+0.67	−0.16	no
nearest-prototype	+0.009 (0.068)	+0.006 (0.136)	+0.63	+0.03	no
mahalanobis	+0.008 (0.085)	+0.006 (0.148)	+0.85	+0.32	no

4.9 Independent replication (BURST)

Every result so far reuses one frozen inference pass on one dataset, so the species and location experiments are near-orthogonal overlays rather than statistically independent replications. To test the null on genuinely independent data we re-ran the whole pipeline on BURST [1]. Its animal subset — the animal LVIS classes of the validation split, streamed mask-native and prompted by class name — yields 132 video $\times$ class cells across 41 species, in a distinct regime (internet/driving/movie video, not camera traps). Zero-shot SAM 3 is again non-degenerate (pHOTA 0.635, Table 4.13), and here the association target is genuine (corr(`pDetA`, `pAssA`) = 0.76; of the 71 multi-object cells, none has `pAssA` exactly equal to `pDetA`), so both components are testable.

A second convergent null. Under leakage-free leave-species-out cross-validation the before-running distances again fail to predict detection transfer: the combined distance model improves out-of-sample MAE by only +0.002 ($p = 0.374$), with visual (+0.003, $p = 0.257$), taxonomic (full-taxonomy subset −0.006, $p = 0.952$) and environment (−0.001, $p = 0.747$) all individually null, and the association specs null likewise (Table 4.14). The result is robust to dropping the 19 single-cell species and to a size control. The lone near-signal, visual distance, is the familiar artefact: its correlation with detection is +0.17 — in the *opposite* direction to the novelty hypothesis (more-novel animals detected better), and a size confound (partial correlation given log-area +0.08, $p = 0.35$). Both datasets — SA-FARI and BURST — now converge on the same before-running distance null.

Table 4.13: Zero-shot SAM 3 on the BURST val animal subset — support-weighted mean over 132 probe cells (41 species), mask-HOTA. The score is non-degenerate and the association target is genuine (corr(`pDetA`, `pAssA`) = 0.76, 71/132 multi-object cells).

Metric (mask-HOTA)	Value
pDetA	0.607
pAssA	0.688
pHOTA	0.635

Convergent, but not proof. At 132 cells BURST supplies a mask-native regime with a genuinely non-degenerate association target, and the distance null holds throughout. It is not, however, positive proof: no before-running quantity — not even the animal-size control that validates on SA-FARI — reaches significance here, so we cannot independently certify that this test could have caught a small species-level effect. The lone near-signal, visual distance, is again the wrong-direction size confound (its correlation with detection is +0.37 between species, the opposite sign to novelty). We therefore read BURST as a second convergent null — no

Table 4.14: BURST independent replication — out-of-sample error under leave-species-out cross-validation (41 species, 132 mask-native cells) vs a mean predictor. ΔMAE is the `pDetA` gain over baseline, with a paired-bootstrap 95% CI and one-sided p. Every distance is null; not even the animal-size control reaches significance, so BURST is a convergent null that cannot certify power for a small species-level effect.

Predictor	ΔMAE	95% CI	p
distances (taxonomic + visual + environment)	+0.002	[−0.007, +0.011]	0.374
visual only	+0.003	[−0.005, +0.013]	0.257
taxonomic only (full-taxonomy subset, n=82)	−0.006	—	0.952
environment only	−0.001	[−0.005, +0.003]	0.747
animal size (log-area) [control]	+0.005	[−0.003, +0.014]	0.143

moderate-or-large before-running distance signal, in an independent mask-native regime with a non-degenerate association target — rather than positive proof, and note that a small species-constant effect cannot be excluded.

4.10 Cross-dataset synthesis

Table 4.15 collects every feature tested, its numerical out-of-sample result, and the dataset it was measured on. The pattern is consistent across both: no before-running label-free distance — taxonomic, visual, or environment, alone or combined — predicts detection or association transfer out of sample; every gain over a mean predictor is within bootstrap noise of zero. The only near-signal, visual distance, appears in the *opposite* direction to the novelty hypothesis (more-novel animals detected better, not worse) and dissolves under an animal-size control wherever it is examined. The only before-running quantity that ever reaches significance is the animal-size confound itself — and only modestly, on the large SA-FARI sample — which doubles as the positive control that keeps the null honest: the same test that stays flat for the distances does pick up size out-of-sample. The contribution is this robust, twice-replicated null, with its bounds stated: consistent across a camera-trap and an internet-video dataset, and powered enough on SA-FARI to exclude a moderate effect, but not a small one.

Table 4.15: Cross-dataset synthesis: every feature tested, its value on each dataset, and the verdict. Each column is reported in one metric, stated in its header: on **BURST** every entry is the out-of-sample ΔMAE (p) over a mean predictor; on **SA-FARI** the individual distances are the standardised GLM coefficient with its 95% CI, while the two rows marked † (combined distances, animal size) are ΔMAE (p) like BURST. A feature "predicts transfer" only if it is positive *and* significant. Only the animal-size confound clears the bar, and only on SA-FARI (the positive control); every label-free distance is null on both datasets.

Feature	SA-FARI (coef. [CI])	BURST (ΔMAE, p)	Verdict
Before-running label-free distances (the research question)			
Combined distances†	−0.004 (0.74)	+0.002 (0.37)	null on both
Visual alone	−0.14 [−0.50, 0.14]	+0.003 (0.26)	null; wrong direction
Taxonomic alone	−0.36 [−3.87, 0.09]	−0.006 (0.95)	null on both
Environment alone	+0.04 [−0.19, 0.24]	−0.001 (0.75)	null on both
Temporal gap	32/1447, all = 0	no timestamps	untested
Confound control			
Animal size (log-area)†	+0.015 (0.01)	+0.005 (0.14)	**validates** (SA-FARI)
Detection precision (hard negatives)			
Hallucination vs. visual	$r = -0.33$; +0.008 (0.053)	—	marginal correlate

4.11 Cross-model replication: the model swap

The two datasets above both vary the data while holding the tracker fixed at SAM 3, so they leave one threat open: the null might be a quirk of SAM 3 rather than a property of the prediction problem. A controlled model swap closes it — hold the cells, the ground truth, the four distances and the GLM fixed, and change only the tracker. If the before-running distances fail to predict a different tracker's transfer too, the null is task-general. We swap in two open-vocabulary, text-promptable trackers of deliberately different architecture: GLEE [30], a single unified detect–segment–track model, and Florence-2 [31] + SAM 2 [24], a two-stage pipeline in which a caption-style generator emits boxes (with no confidence score) and SAM 2 turns them into tracked masks.

SA-FARI cannot host the swap — and it is not the scorer. On SA-FARI both challengers score pDetA ≈ 0 and cannot be modelled at all. This is not a rigged evaluator: the zeros survive box-HOTA (scoring the box models on boxes, GLEE 0.033) and a threshold-free average-precision metric with no confidence gate at all (GLEE 0.000, Florence-2 0.017–0.022); Florence-2, which emits no score, is unaffected by the gate and still scores ~ 0. The cause is a double confound. SA-FARI is SAM 3's own co-released benchmark, and its exotic, nocturnal species are out-of-distribution for internet-trained challengers — and the domain, not merely the species, is what breaks them: run GLEE on SA-FARI's everyday species (jaguar, cow, fox) and it localises them well (oracle IoU up to 0.90) yet its confidence still collapses below any usable operating point on the dark, cluttered camera-trap footage. SA-FARI therefore cannot yield a fair, non-degenerate challenger score.

BURST hosts a fair swap, and both challengers work. BURST is neither SAM 3's benchmark nor camera-trap footage, and SAM 3 scores higher on it (pDetA 0.607) than on its own SA-FARI (0.537) — so there is no home-turf inflation, and the arena is fair. There, both challengers genuinely track: GLEE reaches pDetA 0.340 (median 0.31, spread 0–0.98, 67/132 cells > 0.3) and Florence-2 + SAM 2 reaches 0.394 (median 0.33, 72/132 cells > 0.3), both non-degenerate with the real spread the regression needs (Table 4.16).

Table 4.16: Cross-model swap on BURST (132 cells, 41 species, mask-HOTA): zero-shot detection per tracker — SAM 3 and two architecturally distinct challengers. All three are non-degenerate: both challengers show the real per-cell spread the regression needs (GLEE median 0.31, 67/132 cells > 0.3; Florence-2 + SAM 2 median 0.33, 72/132 cells > 0.3). Their distance-model results are in Table 4.17.

Tracker	mean `pDetA`
SAM 3 (reference)	0.607
GLEE (unified query)	0.340
Florence-2 + SAM 2 (caption)	0.394

The null replicates across models. Under the same leakage-free leave-species-out cross-validation, size-controlled, the combined four-distance model fails to predict either challenger's detection transfer: +0.011 (95% CI [−0.004, +0.027], $p = 0.088$) for GLEE and +0.034 ([−0.001, +0.068], $p = 0.030$) for Florence-2 — both confidence intervals spanning zero, neither clearing the Bonferroni bar ($\alpha = 0.0125$). As on SA-FARI, a few individual distances flicker (for Florence-2, environment $p = 0.006$ and temporal $p = 0.015$), but the discounts are identical: BURST has no locations, so "leave-species-out" is the only valid scheme and an apparent two-scheme pass is one partition counted twice; the flickers route through the animal-size confound the driver controls; and the combined model — what a deployment gate would use — is null in both. Two architecturally distinct trackers thus both (i) genuinely track BURST animals and (ii) show the before-running distance null. **The null is task-general across trackers, not a SAM-3 artefact.**

Table 4.17: Cross-model swap on BURST: out-of-sample error of the combined four-distance model (leave-species-out, size-controlled) vs a mean predictor, per challenger; ΔMAE with a 95% CI and one-sided p (Bonferroni $\alpha = 0.0125$). Both challengers track BURST animals (Table 4.16) yet the distance model is null for each — the label-free distance null replicates across trackers.

Challenger	ΔMAE	95% CI	p
GLEE	+0.011	[−0.004, +0.027]	0.088
Florence-2 + SAM 2	+0.034	[−0.001, +0.068]	0.030

Convergent, but bounded. This is a controlled model swap, not an independent replication: it reuses BURST's cells, ground truth, distances and GLM, changing only the tracker. Its scope is two models on one fair dataset ($n = 132$, underpowered for a small species-level effect, exactly as the SAM 3 BURST run), and Florence-2's combined interval sits closer to zero-exclusion than GLEE's — so we read it as a convergent, confound-free cross-model replication that removes the "single frozen backbone" threat, not as an independently-powered certification. Combined with the two datasets, the null now holds across both axes that could have confounded it — the dataset and the model.

Chapter 5

Discussion

5.1 Interpretation

Neither hypothesis is supported. H1 (detection falls with species novelty) and H2 (association falls with environment difficulty) both fail the pre-registered bar: no label-free distance beats a mean predictor out of sample, and the one coefficient that reached significance — visual distance, running in the opposite direction to the hypothesis (more-novel species detected better, not worse) — is an animal-size artefact. The follow-up intrinsic-difficulty pivot fares no better: low-light, clutter and night/IR predict detection only by proxying animal size, which alone carries the out-of-sample gain. So H0 is what the data support: transfer is *not* distance-driven, and the only label-free property that predicts detection is object size — the very quantity introduced as a nuisance control.

The detection–association decomposition, the intellectual core of the design, has a second, quieter finding: association is a near-degenerate target on this data. Because most cells contain a single object, `pAssA` tracks `pDetA` almost exactly, leaving little identity-specific variance for any feature to predict. The decomposition is therefore a detection story; we report the association null with those divergence figures as its explanation rather than as a separate failed search.

5.1.1 Novelty at the extremes versus graded prediction

A natural objection is that novelty surely matters — a familiar species is detected better than a totally alien one, so how can a novelty distance be null? Both statements are true and do not conflict, because they concern different resolutions of the same axis. The coarse claim — that the extreme tails differ, alien-versus-familiar — is real and never in dispute; the null we report is the far stronger graded claim H1 actually tests: that a continuous distance predicts where on the `pDetA` scale an unseen species lands, out of sample, beyond the size confound, well enough to serve as a deployment gate. These come apart because the informative variance in `pDetA` lives in the middle of the range, not at the poles: the easy extremes (common large animals near 1, hopeless cases near 0) are already predicted by the mean, so a coarse "alien-versus-familiar" signal adds nothing exactly where prediction has value. A genuine extreme-tail effect can therefore coexist with a null graded predictor without contradiction — and it is the graded, held-out, size-controlled predictor, not the truism, that a practitioner would need and that this work shows does not exist. (The model swap sharpens this: the challengers' own detect-or-fail behaviour on exotic species is itself a coarse novelty effect, yet it is that model's novelty, measured against the wrong reference, and it too fails to yield a graded predictor — so it re-derives the hypothesis, not its confirmation.)

5.2 Relation to prior label-free evaluation

The null sits against a literature of positive results, and two axes separate every one of those results from ours — which is why the disagreement is only apparent. The first axis is the signal. Accuracy- and Agreement-on-the-Line [2, 19], ATC [12] and AutoEval [10] all read the model's own behaviour on the target — its confidence, its agreement across a model pair, or (for AutoEval) a feature-distribution distance computed after passing the target images through the network. Those are *after-running* estimators. We instead forecast from external, label-free distances known *before* a single frame is processed — the strictly harder, and for a practitioner deciding whether to spend compute, the more useful regime. The second axis is the output. Every predecessor targets a scalar classifier accuracy [2, 10, 12, 19] or the quality of a static detector/segmenter [28, 33]; none concerns identity maintained over time, and none is decomposed. Ours is the first study to ask the question of a promptable video tracker, split into detection (`pDetA`) and association (`pAssA`). AutoEval [10] is the closest methodological cousin — it, too, regresses a performance score on a distributional feature distance — and the strength of its positive sharpens the contrast: it reports a near-perfect rank correlation (Spearman $\rho \approx -0.91$) between a Fréchet distribution-shift distance and classifier accuracy. That distance, though, is computed from the model's own penultimate features *after* the target images are passed through the network, and predicts a scalar classifier accuracy. We test AutoEval's exact functional in our own regime — the Fréchet and MMD distributional distances on cached DINOv2 embeddings, computed *before* running SAM 3 (§4.7, Table 4.11) — and it stays null ($\Delta\text{MAE} \leq 0.007$, $p \geq 0.10$, every confidence interval spanning zero), so the failure is not an artefact of a lazy nearest-prototype distance: even the strongest AutoEval-style distance carries no before-running transfer signal. The contrast is not merely one of magnitude but of sign: our nearest analogue, the visual distance, correlates with detection in the *opposite* (wrong) direction and dissolves under an animal-size control. Our result therefore does not contradict AutoEval's positive; it extends the recipe to a regime — distance measured before running, target a decomposed video tracker — where the shortcut stops working, and reports that it does. A final bridge makes the boundary concrete: Accuracy-on-the-Line [19] was validated on iWildCam, the same camera-trap shift this dissertation cites via WILDS and Terra Incognita [3, 15] — so the exact wildlife shift where a classifier's OOD accuracy *is* predictable label-free is where our before-running distances are *not*, across both the classifier-to-tracker and after-running-to-before-running boundaries.

5.3 Reconciling the environment null with the background-bias literature

Our environment axis inherits its motivation and its foreground/background separation from a literature that finds backgrounds emphatically *do* matter [3, 5, 25, 32], so the null it returns deserves an explicit reconciliation rather than a bare report. The apparent tension dissolves on two distinctions. The first is **causal reliance versus predictive distance**. Xiao et al. [32], Rosenfeld et al. [25] and PanAf-FGBG [5] all intervene on the background — swap it, transplant the object, or subtract the paired background-only clip's features — and watch performance move; PanAf, most sharply, shows a chimpanzee-behaviour classifier that scores from the background *alone* almost as well as from the foreground on unseen locations, and recovers several points of the location drop by subtracting the background. That is a statement about how a model uses context. Ours is the strictly weaker, correlational question of whether a scalar scene distance-from-reference forecasts an untouched model's per-cell score — and a model can lean causally on context while its score stays unpredictable from a one-number distance. Our null therefore does not deny their causal findings; it locates a regime where the label-free-distance shortcut, not the context reliance, fails.

The second distinction is **why the promptable regime should be the one where it**

fails, and here the same literature supplies the mechanism. Xiao et al. [32], Rosenfeld et al. [25] and PanAf all study models that must infer the category — a classifier with no foreground cue, or a detector reading co-occurrence — for which a spuriously correlated background is a usable shortcut. SAM 3 is told what to find: the prompt `"impala"` supplies the foreground identity, removing the incentive to ride the scene, exactly the direction Xiao et al. [32]'s own finding predicts (better, less-shortcut-prone models close the background gap). PanAf's target compounds this — fine-grained behaviour is intrinsically background-correlated (a chimp nests in trees, drinks at water), whereas ours is finding and following the named animal itself, where naming it short-circuits the context dependence. The result is thus an extension, not a refutation: the very backgrounds that causally drive a camera-trap classifier's location drop do not, once surgically isolated from the animal and used as a before-running distance, forecast a promptable tracker's transfer — and because we isolate the scene from the animal explicitly (a step none of the label-free-prediction methods take), this is a cleaner statement than "background is diagnostic", not a weaker one. The claim is bounded accordingly: it concerns background distance as a predictor of detection transfer for a promptable tracker (the secondary location split, replicated null on BURST, with a near-degenerate association target), not a blanket "background is irrelevant", which would wrongly contradict [3, 5, 32].

5.4 Why label-free distances do not yield a deployment estimator

The field problem that motivated this work — telling a practitioner, before spending compute, whether SAM 3 is trustworthy for a new (species, place) — is not solved by label-free distances. That is a negative but useful answer: the intuition that "far-from-training implies unreliable" does not hold for a promptable video tracker, so a deployment gate cannot be built from taxonomic, visual, environmental or temporal distance to a reference set alone. The one predictive label-free signal, animal size, is too coarse and too obvious to serve as such a gate. What the negative result does establish is a rigorous baseline: any future before-running estimator must clear the same whole-group hold-out bar that these well-motivated distances failed.

One *after-running* signal does clear that bar, and we state it plainly rather than leave it implicit. If a practitioner is willing to run the tracker first, an ATC-style estimator [12] — reading SAM 3's own detection confidence on the target and thresholding it — predicts `pDetA` out of sample on both cross-validation schemes, where every label-free distance failed. We deliberately keep this outside the contribution for three reasons: it is *after-running*, a strictly weaker operating point than the before-running question we set out to answer; it is confirmatory of an established label-free-accuracy family rather than novel; and it is mechanistically close to circular, since its feature and its target are both functions of SAM 3's own output (a cell with no confident masklet trivially has zero detection accuracy). It is a real, validated estimator for the after-running, before-labelling operating point — but not the before-running predictor this dissertation asks for, and not part of the null it reports.

5.5 Limitations and threats to validity

Several limitations bound the claim. First, distances are measured against the SA-FARI train split, *not* SAM 3's true pretraining corpus (Meta's SA-Co), which is undisclosed; the null is scoped to distance-from-SA-FARI. We test one hedge against this — the familiarity proxy (§4.8), which reads separability directly from SAM 3's own features — and it too is null, recovering the same size-entangled visual signal, so even model-internal familiarity does not lift the null; a coverage-anchored proxy for the actual corpus nonetheless remains untested. Second, the analysis is a single dataset and a single frozen backbone, and — since one inference pass is reused across both splits — the species and location experiments are two overlays on the *same* scored cells, not independent replications; they give near-orthogonal but not statistically independent evidence. We take two steps against this. Against the dataset threat we re-run the whole pipeline on an independent dataset: on the larger, mask-native BURST (§4.9, 132 cells / 41 species) the distance null holds again, giving a second convergent null (not positive proof — BURST detects only a large between-species predictor and remains underpowered for a small species-constant distance effect — so we report convergence, not independent confirmation of any positive). We also ran a third dataset, MammAlps, in an earlier pass; it is directionally consistent with the null but we do not count it here, because at roughly 20 cells over 5 species it is too small to certify a null and its per-clip annotations are not reproducible from the public release. We report it as excluded rather than silently drop it. Against the model threat — the "single frozen backbone" — we run the controlled model swap (§4.11): two architecturally distinct open-vocabulary trackers, GLEE and Florence-2 + SAM 2, both track BURST's animals non-degenerately (pDetA 0.34 and 0.39) yet the combined distance model is null for each (CIs span zero, neither clears the Bonferroni bar), so the null is task-general rather than a SAM 3 artefact — a convergent, confound-free cross-model replication (again not an independently-powered certification: two models on one fair dataset). That swap could not run on SA-FARI itself, and for good reason: SA-FARI is SAM 3's own co-released benchmark, and its camera-trap domain collapses the challengers' confidence calibration even where they localise well, so BURST is the fair-swap ceiling. Third, the effective sample is tens of independent species/locations, not hundreds of cells — which is why inference is group-clustered throughout, and why aggregate correlations (e.g. the location-level night/IR effect) are reported alongside their much weaker cell-level values to avoid the ecological-correlation fallacy. Representational probing is only partly opened here: the familiarity proxy (§4.8) probes whether SAM 3's features separate the species — they do, but that separability merely re-encodes the visual distance and does not predict transfer — while a fuller probe of what those representations encode remains open.

Finally, on the scorer itself: all scores come from the unmodified official SA-FARI evaluator, which wraps the standard HOTA/TETA tracking metrics rather than any SAM-3-specific logic. Its one SAM-3-calibrated knob is a fixed 0.5 confidence gate; we verified it is not load-bearing for the cross-model comparison, since the challengers' near-zero SA-FARI scores survive both a box-based metric and a threshold-free average-precision metric with no gate at all (§4.11). The measurement is therefore model-fair, and the challenger failures on SA-FARI are genuine detection failures, not scoring artefacts.

5.6 Ethical considerations

The data are anonymised camera-trap footage released under a non-commercial licence; location identifiers are already anonymised in SA-FARI. The negative result carries a responsible-deployment message: because label-free distance does *not* reliably flag when SAM 3 will fail, conservation teams should not treat low-distance-from-training as a safety guarantee, and reliability should be checked with labelled spot-audits rather than assumed.

Chapter 6

Conclusion

6.1 Summary of findings

We asked whether a promptable tracker's zero-shot accuracy on an unseen (species, place) — studied primarily on SAM 3 — can be predicted, before running the model and without scoring its output, from four label-free distances — separately for detection and association. Under a deliberately powerful out-of-sample test the answer is no: no distance beats a mean-predictor baseline on either a species or a location hold-out, the one apparent effect is an animal-size confound, and a follow-up intrinsic-difficulty pivot collapses to the same size effect. Only object size predicts detection, modestly, and it is the nuisance the analysis controls for. Association carries almost no signal distinct from detection. The null is robust to the two ways it might have been an artefact: it holds across two datasets (SA-FARI and BURST), and, in a controlled model swap on BURST, across two architecturally distinct promptable trackers, so it reflects the prediction problem rather than any one dataset or model. The result is a rigorous, decomposed, and now cross-model null.

6.2 Contributions

To our knowledge this is the first study to test, out of sample, whether a promptable video tracker's zero-shot transfer is predictable from label-free, before-running signals, and the first to decompose that question into detection and association. Its contribution is the honest negative answer and its explanation: transfer is not distance-driven; the apparent signals reduce to an animal-size confound; the test is shown to detect that one real signal when present, so the null reflects an absence of signal, not of power; and the null generalises — across two datasets and, via the controlled model swap, across two other trackers — so it is task-general rather than specific to SAM 3.

6.3 Future work

Three directions remain open. First, a fuller representational probe. The familiarity proxy already tests whether SAM 3's own features separate the species and finds they do, but only by re-encoding the visual distance (Section 4.8); what those features actually encode — whether taxonomy is represented structurally, and a coverage-anchored proxy for the true, undisclosed pretraining corpus — is left untested. Second, broadening the model swap: more trackers, and ideally a larger fair, mask-native dataset than BURST that is not any tracker's home turf — SA-FARI's larger cell count could not be reused, because its camera-trap domain collapses internet-trained challengers' calibration even where they localise well, making it the fair-swap ceiling here. Third, targets with richer association structure (crowded, multi-object scenes) where the detection–association decomposition would have more to distinguish.

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